having row index c).

By line 14, we know that if $0 \le c < numCols-1$, then initially, before the 1st iteration of the loop of lines 15-19, we get:

$\boxed{startingPos[c+1] = startingPos[c] + rowTerms[c]}$

- This means, initially $(rowTerms[c]-1)$ indices of matrix b[] in between startingPos[c] and startingPos[c+1] are left unmarked.
- Let, for all indices K, $1 \le K < x$, let $a[K].col = c$ hold for K_1 no. of times. ~~By our above justification, we now~~ know just before the start of the loop iteration when $i = x$, current_startingPos[c] = initial_startingPos[c] + $K_1 <$ initial_startingPos[c+1], since $\boxed{K < rowTerms[c]}$ must hold. This holds since by the loop invariant, all the entries of b[] in between the indices initial_startingPos[c] and current_startingPos[c]−1 have been correctly filled, and on line 16 by postfix operation, we continuously increase by 1 the value of startingPos[c].
- By correctness of loop in lines 13-14, we know $\forall j', 0 \le j' < numCols$, the terms in b[] having row index j' (or column index j' in a[]) will be stored in between the indices startingPos[j'] and startingPos[j'+1] (non-including), where we have assumed startingPos[numCols] is the starting index beginning from which no entry will be ever filled.

∴ We know, current_startingPos[c] is empty. We have initialized to j its value, and increased the index no. of current_startingPos[c] by 1 on line 16, which is natural.

(by defn. of transpose mat)

- The correct transposed entry of the entry at a[x] is $\langle a[x].col, a[x].row, a[x].value \rangle$. ∴ $b[j].row = a[x].col$, $b[j].col = a[x].row$ and $b[j].value = a[x].value$. ∴ We have correctly initialized these values on lines 17 and 18.